

week4_Introduction to tidyverse

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First, we need to indicate where is located the file

```
FilePath <- file.path("C:/Users/Macarena/CytometryInR/course/week04/04_IntroToTidyverse/data",  
FilePath
```

```
library(utils)
```

```
Data <- read.csv(file=FilePath, check.names=FALSE)  
colnames(Data)
```

```
[1] "bid" "timepoint" "Condition" "Date" "infant_sex" "ptype"  
[7] "root" "singletsFSC" "singletsSSC" "singletsSSCB" "CD45" "NotMonocytes"  
[13] "nonDebris" "lymphocytes" "live" "Dump+" "Dump-" "Tcells"  
[19] "Vd2+" "Vd2-" "Va7.2+" "Va7.2-" "CD4+" "CD4-"  
[25] "CD8+" "CD8-" "Tcells_count" "lymphocytes_count" "Monocytes" "Debris"  
[31] "CD45_count"
```

when we put check.names into TRUE, then the "+" or "-" symbols dissappear

```
Data <- read.csv(file=FilePath, check.names=TRUE)  
colnames(Data)
```

```
[1] "bid" "timepoint" "Condition" "Date" "infant_sex" "ptype"  
[7] "root" "singletsFSC" "singletsSSC" "singletsSSCB" "CD45" "NotMonocytes"  
[13] "nonDebris" "lymphocytes" "live" "Dump." "Dump..1" "Tcells"  
[19] "Vd2." "Vd2..1" "Va7.2." "Va7.2..1" "CD4." "CD4..1"  
[25] "CD8." "CD8..1" "Tcells_count" "lymphocytes_count" "Monocytes" "Debris"  
[31] "CD45_count"
```

To see some rows:

```
head(Data, 3)
```

```
bid timepoint Condition Date infant_sex ptype root singletsFSC singletsSSC singletsSSCB
CD45 NotMonocytes nonDebris 1 INF0052 0 Ctrl 2025-07-26 Male HEU-hi 2098368 1894070
1666179 1537396 0.5952943 0.8820349 0.8627649 2 INF0100 0 Ctrl 2025-07-26 Male HEU-lo
2020184 1791890 1697083 1579098 0.9106762 0.9052256 0.8602660 3 INF0100 4 Ctrl 2025-07-
26 Male HEU-lo 1155040 1033320 875465 845446 0.9705765 0.9845400 0.9578793 lymphocytes
live Dump+ Dump- Tcells Vd2+ Vd2- Va7.2+ Va7.2- CD4+ CD4- CD8+ 1 0.6420138
0.9020581 0.21090996 0.6911482 0.2804264 0.008120361 0.9918796 0.01448070 0.9773989
0.6341164 0.3432825 0.2734826 2 0.2145848 0.8908981 0.06252775 0.8283703 0.6748298
0.007265620 0.9927344 0.01577499 0.9769594 0.6119112 0.3650482 0.3357696 3 0.7403110
0.8757665 0.20023803 0.6755285 0.6119129 0.004651313 0.9953487 0.01579402 0.9795547
0.6639621 0.3155925 0.2862104 CD8- Tcells_count lymphocytes_count Monocytes Debris
CD45_count 1 0.06979990 164771 587573 0.11796509 0.13723513 915203 2 0.02927858 208241
308583 0.09477437 0.13973396 1438047 3 0.02938209 371723 607477 0.01545999 0.04212072
820570
```

To visualize the table, we can click into the grid, in the right panel, or using the next code:

```
View(Data)
```

To see the type of data:

```
class(Data)
```

```
[1] "data.frame"
```

And see the variables:

```
str(Data)
```

```
‘data.frame’: 196 obs. of 31 variables: $ bid : chr "INF0052" "INF0100" "INF0100" "INF0100"
... $ timepoint : int 0 0 4 9 0 4 9 0 4 9 0 ... $ Condition : chr "Ctrl" "Ctrl" "Ctrl" "Ctrl" ... $ Date :
chr "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" ... $ infant_sex : chr "Male" "Male"
"Male" "Male" ... $ ptype : chr "HEU-hi" "HEU-lo" "HEU-lo" "HEU-lo" ... $ root : int 2098368
2020184 1155040 358624 1362216 1044808 1434840 972056 1521928 2363512 ... $ singletsFSC
: int 1894070 1791890 1033320 328624 1206309 917398 1265022 875707 1359574 2136616 ...
$ singletsSSC : int 1666179 1697083 875465 289327 1032946 735579 988445 767323 1175755
1875394 ... $ singletsSSCB : int 1537396 1579098 845446 276289 982736 685592 940454 718000
1097478 1732620 ... $ CD45 : num 0.595 0.911 0.971 0.982 0.957 ... $ NotMonocytes : num 0.882
0.905 0.985 0.986 0.956 ... $ nonDebris : num 0.863 0.86 0.958 0.941 0.841 ... $ lymphocytes :
num 0.642 0.215 0.74 0.651 0.705 ... $ live : num 0.902 0.891 0.876 0.915 0.895 ... $ Dump+
```

```
: num 0.2109 0.0625 0.2002 0.2147 0.3383 ... $ Dump- : num 0.691 0.828 0.676 0.701 0.557 ...
$ Tcells : num 0.28 0.675 0.612 0.631 0.44 ... $ Vd2+ : num 0.00812 0.00727 0.00465 0.01135
0.00475 ... $ Vd2- : num 0.992 0.993 0.995 0.989 0.995 ... $ Va7.2+ : num 0.0145 0.0158 0.0158
0.017 0.0133 ... $ Va7.2- : num 0.977 0.977 0.98 0.972 0.982 ... $ CD4+ : num 0.634 0.612 0.664
0.438 0.739 ... $ CD4- : num 0.343 0.365 0.316 0.534 0.243 ... $ CD8+ : num 0.273 0.336 0.286
0.486 0.195 ... $ CD8- : num 0.0698 0.0293 0.0294 0.0476 0.0476 ... $ Tcells_count : int 164771
208241 371723 111552 291777 271870 487937 220634 415867 184930 ... $ lymphocytes_count:
int 587573 308583 607477 176662 663667 510730 726238 451047 710964 652155 ... $ Monocytes
: num 0.118 0.0948 0.0155 0.0145 0.0444 ... $ Debris : num 0.1372 0.1397 0.0421 0.0587 0.1592
... $ CD45_count : int 915203 1438047 820570 271304 940733 675857 921660 701657 1066884
1017713 ...
```

To see that using glimpse function (if we don't know from what package is, we can use the next code to figure out)

```
::: {.cell}
```

```
??glimpse
```

```
:::
```

It's from dplyr package, so we have to activate it

```
::: {.cell}
```

```
library(dplyr)
glimpse(Data)
```

```
:::
```

```
Rows: 196 Columns: 31 $ bid "INF0052", "INF0100", "INF0100", "INF0100", "INF0179",
"INF0179", "INF0179", "INF0186", "INF0186", "INF0052", ... $ timepoint 0, 0, 4, 9, 0,
4, 9, 4, 9, 0, 0, 4, 9, 0, 4, 9, 4, 9, 0, 0, 4, 0, 4, 9, 4, 9, 0, 4, 9, 0, 4, 9,...
$ Condition "Ctrl", "Ctrl", "Ctrl", "Ctrl", "Ctrl", "Ctrl", "Ctrl", "Ctrl", "Ctrl", "PPD",
"PPD", "PPD", "PPD", "PPD", "PP... $ Date "2025-07-26", "2025-07-26", "2025-07-26",
"2025-07-26", "2025-07-26", "2025-07-26", "2025-07-26"... $ infant_sex "Male",
"Male", "Male", "Male", "Male", "Male", "Male", "Female", "Female", "Male", "Male",
"Male", "Male", "M... $ ptype "HEU-hi", "HEU-lo", "HEU-lo", "HEU-lo", "HU", "HU",
"HU", "HEU-hi", "HEU-hi", "HEU-hi", "HEU-lo", "HEU-lo", "H... $ root 2098368, 2020184,
1155040, 358624, 1362216, 1044808, 1434840, 972056, 1521928, 2363512, 2049112, 1063496,
7883... $ singletsFSC 1894070, 1791890, 1033320, 328624, 1206309, 917398, 1265022, 875707,
1359574, 2136616, 1821676, 946587, 714198... $ singletsSSC 1666179, 1697083, 875465,
289327, 1032946, 735579, 988445, 767323, 1175755, 1875394, 1717636, 796056, 626387, ... $
singletsSSCB 1537396, 1579098, 845446, 276289, 982736, 685592, 940454, 718000, 1097478,
```

```

1732620, 1597085, 767297, 600011, 1... $ CD45 0.5952943, 0.9106762, 0.9705765, 0.9819573,
0.9572591, 0.9858006, 0.9800160, 0.9772382, 0.9721234, 0.5873838, ... $ NotMonocytes
0.8820349, 0.9052256, 0.9845400, 0.9855070, 0.9556272, 0.9622894, 0.9805210, 0.9490663,
0.9581698, 0.8619837, ... $ nonDebris 0.8627649, 0.8602660, 0.9578793, 0.9412615, 0.8407837,
0.9499243, 0.9657759, 0.9189704, 0.9186763, 0.8429685, ... $ lymphocytes 0.6420138, 0.2145848,
0.7403110, 0.6511588, 0.7054786, 0.7556776, 0.7879674, 0.6428312, 0.6663930, 0.6408044,
... $ live 0.9020581, 0.8908981, 0.8757665, 0.9153242, 0.8952140, 0.9673056, 0.9297420,
0.8756515, 0.8749529, 0.9009254, ... $ Dump+ 0.21090996, 0.06252775, 0.20023803, 0.21469246,
0.33831877, 0.36947311, 0.23863114, 0.32516789, 0.22875701, 0.... $ Dump- 0.6911482,
0.8283703, 0.6755285, 0.7006317, 0.5568953, 0.5978325, 0.6911109, 0.5504837, 0.6461959,
0.6934931, ... $ Tcells 0.2804264, 0.6748298, 0.6119129, 0.6314431, 0.4396437, 0.5323165,
0.6718693, 0.4891597, 0.5849340, 0.2835676, ... $ Vd2+ 0.008120361, 0.007265620, 0.004651313,
0.011348967, 0.004753630, 0.005160555, 0.002631487, 0.009300470, 0.0179... $ Vd2- 0.9918796,
0.9927344, 0.9953487, 0.9886510, 0.9952464, 0.9948394, 0.9973685, 0.9906995, 0.9820592,
0.9925918, ... $ Va7.2+ 0.014480704, 0.015774991, 0.015794019, 0.017023451, 0.013321818,
0.012461838, 0.013563226, 0.038770090, 0.0367... $ Va7.2- 0.9773989, 0.9769594, 0.9795547,
0.9716276, 0.9819246, 0.9823776, 0.9838053, 0.9519294, 0.9452878, 0.9775212, ... $ CD4+
0.6341164, 0.6119112, 0.6639621, 0.4378944, 0.7392563, 0.5876264, 0.6994141, 0.5724730,
0.5119209, 0.6340345, ... $ CD4- 0.3432825, 0.3650482, 0.3155925, 0.5337331, 0.2426682,
0.3947512, 0.2843912, 0.3794565, 0.4333669, 0.3434867, ... $ CD8+ 0.2734826, 0.3357696,
0.2862104, 0.4861231, 0.1950634, 0.3571376, 0.2580784, 0.3555526, 0.4138799, 0.2744119, ... $
CD8- 0.06979990, 0.02927858, 0.02938209, 0.04761008, 0.04760485, 0.03761357, 0.02631282,
0.02390384, 0.01948700, 0.... $ Tcells_count 164771, 208241, 371723, 111552, 291777, 271870,
487937, 220634, 415867, 184930, 211987, 326378, 238021, 294549... $ lymphocytes_count
587573, 308583, 607477, 176662, 663667, 510730, 726238, 451047, 710964, 652155, 314717,
544883, 366784, 663169... $ Monocytes 0.11796509, 0.09477437, 0.01545999, 0.01449297,
0.04437285, 0.03771064, 0.01947898, 0.05093372, 0.04183023, 0.... $ Debris 0.13723513,
0.13973396, 0.04212072, 0.05873854, 0.15921627, 0.05007568, 0.03422412, 0.08102962,
0.08132374, 0.... $ CD45_count 915203, 1438047, 820570, 271304, 940733, 675857, 921660,
701657, 1066884, 1017713, 1447451, 745037, 589379

```

This function is similar to `str()` function but more tidy and with the dimensions of the table at the top. To see only the dimensions, we can also use `dim()` function, which maintains the row followed by column position convention of base R (first row, second column)

```
dim(Data)
```

[1] 196 31

To verify the column value type (character, for instance), we can use the functions `is.character`, `is.numeric` or `is.logical`:

```
is.character(Data$bid)
```

```
[1] TRUE
```

```
is.numeric(Data$bid)
```

```
[1] FALSE
```

```
is.character(Data$Tcells_count)
```

[1] FALSE For numeric columns using the `is.numeric()` function, we can also be subtype specific using either `is.integer()` or `is.double()`:

```
is.numeric(Data$Tcells_count)
```

```
[1] TRUE
```

```
is.integer(Data$Tcells_count)
```

```
[1] TRUE
```

```
is.double(Data$Tcells_count)
```

```
[1] FALSE
```

Like with keywords, special characters (as `&` or spaces) will need tick marks to run the function

```
is.numeric(Data$`CD8-`)
```

```
[1] TRUE
```

```
library(dplyr)
```

The `select()` function is used to “select” a column from a `data.frame` type object. The simplest way to use it is giving the name of the `data.frame` object in first place and then, the name of the column, surrounded by ” “:

```
DateColumnn <- select(Data, "Date")
DateColumnn[1:10,]
```

```
[1] "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26"
```

This results in the column being selected, resulting in the new object containing only that subsetting out column from the original Data object.

In general, the function `select()` is used with the implementation of pipe operators, not like we used previously:

```
DateColumnn <- Data |> select("Date")
DateColumnn[1:10,]
```

```
[1] "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26"
```

The main advantage of the pipe operators is that we can link different functions. For example, the `nrow()` function determine the number of rows. We can link the function `select()` and `nrow()` with the pipe operators:

```
::: {.cell}
```

```
Data |> select("Date") |> nrow()
```

```
:::
```

```
[1] 196
```

CURIOUS NOTE: Despite we need “ ” for `install.packages()` function, for instance, in `select()` function, the function still run without the “ ” in date:

```
::: {.cell}
```

```
Data |> select(Date) |> head(3)
```

```
:::
```

```
  Date
```

```
1 2025-07-26 2 2025-07-26 3 2025-07-26
```

This is due to dplyr; It's now in charge and it's able to infer that Date is a column, with no need of “”. For select more than one column and position them anywhere within the data.frame:

```
Subset <- Data |> select(bid, timepoint, Condition, Tcells, `CD8+`, `CD4+`)
head(Subset, 3)
```

```
      bid timepoint Condition      Tcells      CD8+      CD4+
1 INF0052 0 Ctrl 0.2804264 0.2734826 0.6341164 2 INF0100 0 Ctrl 0.6748298 0.3357696
0.6119112 3 INF0100 4 Ctrl 0.6119129 0.2862104 0.6639621
```

The order in which we selected the columns will dictate their position in the subsetted data.frame object:

```
Subset <- Data |> select(bid, Tcells, `CD8+`, `CD4+`, timepoint, Condition)
head(Subset, 3)
```

```
      bid      Tcells      CD8+      CD4+ timepoint Condition
1 INF0052 0.2804264 0.2734826 0.6341164 0 Ctrl 2 INF0100 0.6748298 0.3357696 0.6119112 0
Ctrl 3 INF0100 0.6119129 0.2862104 0.6639621 4 Ctrl
```

The relocate() function is used when we want to move just one column. We can do this with select() function too, but it would be a waste of time. Let's say we wanted to move the Tcells column from its current location to the second column position (right after the bid column). The line of code to do so would look like:

```
Data |> relocate(Tcells, .after=bid) |> head(3)
```

```
      bid      Tcells timepoint Condition      Date infant_sex ptype      root singletsFSC singlets
1 INF0052 0.2804264 0 Ctrl 2025-07-26 Male HEU-hi 2098368 1894070 1666179 1537396
0.5952943 0.8820349 2 INF0100 0.6748298 0 Ctrl 2025-07-26 Male HEU-lo 2020184 1791890
1697083 1579098 0.9106762 0.9052256 3 INF0100 0.6119129 4 Ctrl 2025-07-26 Male
HEU-lo 1155040 1033320 875465 845446 0.9705765 0.9845400 nonDebris lymphocytes live
Dump+ Dump- Vd2+ Vd2- Va7.2+ Va7.2- CD4+ CD4- CD8+ 1 0.8627649 0.6420138
0.9020581 0.21090996 0.6911482 0.008120361 0.9918796 0.01448070 0.9773989 0.6341164
0.3432825 0.2734826 2 0.8602660 0.2145848 0.8908981 0.06252775 0.8283703 0.007265620
0.9927344 0.01577499 0.9769594 0.6119112 0.3650482 0.3357696 3 0.9578793 0.7403110
```

```
0.8757665 0.20023803 0.6755285 0.004651313 0.9953487 0.01579402 0.9795547 0.6639621
0.3155925 0.2862104 CD8- Tcells_count lymphocytes_count Monocytes Debris CD45_count
1 0.06979990 164771 587573 0.11796509 0.13723513 915203 2 0.02927858 208241 308583
0.09477437 0.13973396 1438047 3 0.02938209 371723 607477 0.01545999 0.04212072 820570
```

Similar to what we saw with `select()`, this approach can also be used for more than 1 column:

```
Data |> relocate(Tcells, Monocytes, .after=bid) |> head(3)
```

	bid	Tcells	Monocytes	timepoint	Condition	Date	infant_sex	ptype	root	singlets	
1	INF0052	0.2804264	0.11796509	0	Ctrl	2025-07-26	Male	HEU-hi	2098368	1894070	1666179
1537396	0.5952943	2	INF0100	0.6748298	0.09477437	0	Ctrl	2025-07-26	Male	HEU-lo	2020184
1791890	1697083	1579098	0.9106762	3	INF0100	0.6119129	0.01545999	4	Ctrl	2025-07-26	Male
HEU-lo	1155040	1033320	875465	845446	0.9705765	NotMonocytes	nonDebris	lympho-			
cytes	live	Dump+	Dump-	Vd2+	Vd2-	Va7.2+	Va7.2-	CD4+	CD4-	1	0.8820349
	0.6420138	0.9020581	0.21090996	0.6911482	0.008120361	0.9918796	0.01448070	0.9773989			
	0.6341164	0.3432825	2	0.9052256	0.8602660	0.2145848	0.8908981	0.06252775	0.8283703		
	0.007265620	0.9927344	0.01577499	0.9769594	0.6119112	0.3650482	3	0.9845400	0.9578793		
	0.7403110	0.8757665	0.20023803	0.6755285	0.004651313	0.9953487	0.01579402	0.9795547			
	0.6639621	0.3155925	CD8+	CD8-	Tcells_count	lymphocytes_count	Debris	CD45_count			
1	0.2734826	0.06979990	164771	587573	0.13723513	915203	2	0.3357696	0.02927858	208241	308583
	0.13973396	1438047	3	0.2862104	0.02938209	371723	607477	0.04212072	820570		

To rename columns, we can proceed calling a subset of data and then, specifying the column (third, first, whatever). Once we confirm the column that we want to rename, next we can change it:

```
colnames(Subset)
```

```
[1] "bid" "Tcells" "CD8+" "CD4+" "timepoint" "Condition"
```

Now, we specify the column in which we are interested:

```
colnames(Subset)[3]
```

And we change the name of that particular column:

```
colnames(Subset)[3] <- "CD8Positive"
colnames(Subset)
```

```
[1] "bid" "Tcells" "CD8Positive" "CD4+" "timepoint" "Condition"
```

We can also do that in a different way:


```
Renamed <- Subset |> rename(CD4_Positive = `CD4+`)
colnames(Renamed)
```

```
[1] "bid" "Tcells" "CD8Positive" "CD4_Positive" "timepoint" "Condition"
```

If we want to rename several columns at once, we can do it using a comma between them:

```
Renamed_Multiple <- Subset |> rename(specimen = bid, timepoint_months = timepoint, stimulation = Condition)
colnames(Renamed_Multiple)
```

```
[1] "specimen" "Tcells" "CD8Positive" "CD4_Positive" "timepoint_months" "stimulation"
```

To see data from a particular column, we use the function `pull()`:

```
Data |> pull(Date) |> head(5)
```

```
[1] "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26" "2025-07-26"
```

If we also want to see the unique values in a column, we can use the `unique()` function:

```
Data |> pull(Date) |> unique()
```

```
[1] "2025-07-26" "2025-07-29" "2025-07-31" "2025-08-05" "2025-08-07" "2025-08-22" "2025-08-28" "2025-08-30"
```

The Condition column in this Dataset appears to be indicating whether the samples were stimulated. Let's see how many unique values are contained within that column

```
Data |> pull(Condition) |> unique()
```

```
[1] "Ctrl" "PPD" "SEB"
```

If we want to subset just one treatment, we can use the function `filter()`: IMPORTANT NOTE: In this case, “ ” are necessary as we are asking for a matching character value.

```
PPDOnly <- Data |> filter(Condition == "PPD")
head(PPDOnly, 5)
```

```
bid timepoint Condition      Date infant_sex  ptype      root singletsFSC singletsSSC singletsSSC
```

```

1 INF0052 0 PPD 2025-07-26 Male HEU-hi 2363512 2136616 1875394 1732620 0.5873838
0.8619837 0.8429685 2 INF0100 0 PPD 2025-07-26 Male HEU-lo 2049112 1821676 1717636
1597085 0.9063081 0.9251961 0.8771889 3 INF0100 4 PPD 2025-07-26 Male HEU-lo 1063496
946587 796056 767297 0.9709891 0.9848719 0.9556049 4 INF0100 9 PPD 2025-07-26 Male
HEU-lo 788368 714198 626387 600011 0.9822803 0.9842139 0.8123041 5 INF0179 0 PPD
2025-07-26 Male HU 1380336 1242311 1047081 1000877 0.9470275 0.9575685 0.9134438
lymphocytes live Dump+ Dump- Tcells Vd2+ Vd2- Va7.2+ Va7.2- CD4+ CD4- CD8+ 1
0.6408044 0.9009254 0.20743228 0.6934931 0.2835676 0.007408209 0.9925918 0.01507057
0.9775212 0.6340345 0.3434867 0.2744119 2 0.2174284 0.8929673 0.06181426 0.8311531
0.6735798 0.007137230 0.9928628 0.01671801 0.9761448 0.6145707 0.3615741 0.3312279 3
0.7313503 0.8782307 0.20727202 0.6709587 0.5989873 0.005254643 0.9947454 0.01609790
0.9786475 0.6559480 0.3226994 0.2912084 4 0.6223228 0.9566639 0.23164587 0.7250180
0.6489405 0.011935922 0.9880641 0.01855298 0.9695111 0.4306889 0.5388222 0.4908558 5
0.6996502 0.8856898 0.33186111 0.5538287 0.4441538 0.004382972 0.9956170 0.01297237
0.9826447 0.7499194 0.2327253 0.1850897 CD8- Tcells_count lymphocytes_count Mono-
cytes Debris CD45_count 1 0.06907479 184930 652155 0.13801632 0.15703150 1017713 2
0.03034620 211987 314717 0.07480391 0.12281107 1447451 3 0.03149109 326378 544883
0.01512811 0.04439511 745037 4 0.04796636 238021 366784 0.01578611 0.18769586 589379 5
0.04763554 294549 663169 0.04243146 0.08655621 947858

```

Although the code previously used works, it can generate error, specially with character values. For that reason, it is better to use `%in%` operator to extract rows from a column.

```
Data |> filter(Condition %in% "PPD") |> head(5)
```

bid	timepoint	Condition	Date	infant_sex	ptype	root	singletsFSC	singletsSSC	singl		
1	INF0052	0 PPD	2025-07-26	Male	HEU-hi	2363512	2136616	1875394	1732620	0.5873838	
0.8619837	0.8429685	2	INF0100	0 PPD	2025-07-26	Male	HEU-lo	2049112	1821676	1717636	
1597085	0.9063081	0.9251961	0.8771889	3	INF0100	4 PPD	2025-07-26	Male	HEU-lo	1063496	
946587	796056	767297	0.9709891	0.9848719	0.9556049	4	INF0100	9 PPD	2025-07-26	Male	
HEU-lo	788368	714198	626387	600011	0.9822803	0.9842139	0.8123041	5	INF0179	0 PPD	
2025-07-26	Male	HU	1380336	1242311	1047081	1000877	0.9470275	0.9575685	0.9134438		
lymphocytes	live	Dump+	Dump-	Tcells	Vd2+	Vd2-	Va7.2+	Va7.2-	CD4+	CD4-	CD8+
1	0.6408044	0.9009254	0.20743228	0.6934931	0.2835676	0.007408209	0.9925918	0.01507057			
0.9775212	0.6340345	0.3434867	0.2744119	2	0.2174284	0.8929673	0.06181426	0.8311531			
0.6735798	0.007137230	0.9928628	0.01671801	0.9761448	0.6145707	0.3615741	0.3312279	3			
0.7313503	0.8782307	0.20727202	0.6709587	0.5989873	0.005254643	0.9947454	0.01609790				
0.9786475	0.6559480	0.3226994	0.2912084	4	0.6223228	0.9566639	0.23164587	0.7250180			
0.6489405	0.011935922	0.9880641	0.01855298	0.9695111	0.4306889	0.5388222	0.4908558	5			
0.6996502	0.8856898	0.33186111	0.5538287	0.4441538	0.004382972	0.9956170	0.01297237				
0.9826447	0.7499194	0.2327253	0.1850897	CD8-	Tcells_count	lymphocytes_count	Mono-	cytes	Debris	CD45_count	
1	0.06907479	184930	652155	0.13801632	0.15703150	1017713	2				

```
0.03034620 211987 314717 0.07480391 0.12281107 1447451 3 0.03149109 326378 544883
0.01512811 0.04439511 745037 4 0.04796636 238021 366784 0.01578611 0.18769586 589379 5
0.04763554 294549 663169 0.04243146 0.08655621 947858
```

As we saw in previous functions, we can also filter several rows or conditions at once:

```
Data |> filter(Condition %in% c("PPD", "SEB")) |> head(5)
```

bid	timepoint	Condition	Date	infant_sex	ptype	root	singletsFSC	singletsSSC	singlets
1	INF0052 0	PPD	2025-07-26	Male	HEU-hi	2363512 2136616 1875394 1732620	0.5873838		
0.8619837	0.8429685	2	INF0100 0	PPD	2025-07-26	Male	HEU-lo	2049112 1821676 1717636	
1597085	0.9063081	0.9251961	0.8771889	3	INF0100 4	PPD	2025-07-26	Male	HEU-lo
946587	796056	767297	0.9709891	0.9848719	0.9556049	4	INF0100 9	PPD	2025-07-26
Male	HEU-lo	788368	714198	626387	600011	0.9822803	0.9842139	0.8123041	5
INF0179 0	PPD	2025-07-26	Male	HU	1380336	1242311	1047081	1000877	0.9470275
0.9575685	0.9134438	lymphocytes	live	Dump+	Dump-	Tcells	Vd2+	Vd2-	Va7.2+
Va7.2-	CD4+	CD4-	CD8+	1	0.6408044	0.9009254	0.20743228	0.6934931	0.2835676
0.007408209	0.9925918	0.01507057	0.9775212	0.6340345	0.3434867	0.2744119	2	0.2174284	0.8929673
0.06181426	0.8311531	0.6735798	0.007137230	0.9928628	0.01671801	0.9761448	0.6145707	0.3615741	0.3312279
3	0.7313503	0.8782307	0.20727202	0.6709587	0.5989873	0.005254643	0.9947454	0.01609790	
0.9786475	0.6559480	0.3226994	0.2912084	4	0.6223228	0.9566639	0.23164587	0.7250180	
0.6489405	0.011935922	0.9880641	0.01855298	0.9695111	0.4306889	0.5388222	0.4908558	5	
0.6996502	0.8856898	0.33186111	0.5538287	0.4441538	0.004382972	0.9956170	0.01297237		
0.9826447	0.7499194	0.2327253	0.1850897	CD8-	Tcells_count	lymphocytes_count	Mono-	cytes	Debris
CD45_count	1	0.06907479	184930	652155	0.13801632	0.15703150	1017713	2	
0.03034620	211987	314717	0.07480391	0.12281107	1447451	3	0.03149109	326378	544883
0.01512811	0.04439511	745037	4	0.04796636	238021	366784	0.01578611	0.18769586	589379
5	0.04763554	294549	663169	0.04243146	0.08655621	947858			

Or we can set up the vector first and then filter the vector:

```
TheseConditions <- c("PPD", "SEB")
Data |> filter(Condition %in% TheseConditions) |> head(5)
```

bid	timepoint	Condition	Date	infant_sex	ptype	root	singletsFSC	singletsSSC	singlets
1	INF0052 0	PPD	2025-07-26	Male	HEU-hi	2363512 2136616 1875394 1732620	0.5873838		
0.8619837	0.8429685	2	INF0100 0	PPD	2025-07-26	Male	HEU-lo	2049112 1821676 1717636	
1597085	0.9063081	0.9251961	0.8771889	3	INF0100 4	PPD	2025-07-26	Male	HEU-lo
946587	796056	767297	0.9709891	0.9848719	0.9556049	4	INF0100 9	PPD	2025-07-26
Male	HEU-lo	788368	714198	626387	600011	0.9822803	0.9842139	0.8123041	5
INF0179 0	PPD								

```

2025-07-26 Male HU 1380336 1242311 1047081 1000877 0.9470275 0.9575685 0.9134438
lymphocytes live Dump+ Dump- Tcells Vd2+ Vd2- Va7.2+ Va7.2- CD4+ CD4- CD8+ 1
0.6408044 0.9009254 0.20743228 0.6934931 0.2835676 0.007408209 0.9925918 0.01507057
0.9775212 0.6340345 0.3434867 0.2744119 2 0.2174284 0.8929673 0.06181426 0.8311531
0.6735798 0.007137230 0.9928628 0.01671801 0.9761448 0.6145707 0.3615741 0.3312279 3
0.7313503 0.8782307 0.20727202 0.6709587 0.5989873 0.005254643 0.9947454 0.01609790
0.9786475 0.6559480 0.3226994 0.2912084 4 0.6223228 0.9566639 0.23164587 0.7250180
0.6489405 0.011935922 0.9880641 0.01855298 0.9695111 0.4306889 0.5388222 0.4908558 5
0.6996502 0.8856898 0.33186111 0.5538287 0.4441538 0.004382972 0.9956170 0.01297237
0.9826447 0.7499194 0.2327253 0.1850897 CD8- Tcells_count lymphocytes_count Mono-
cytes Debris CD45_count 1 0.06907479 184930 652155 0.13801632 0.15703150 1017713 2
0.03034620 211987 314717 0.07480391 0.12281107 1447451 3 0.03149109 326378 544883
0.01512811 0.04439511 745037 4 0.04796636 238021 366784 0.01578611 0.18769586 589379 5
0.04763554 294549 663169 0.04243146 0.08655621 947858

```

This is OK for a low number of conditions, but when we want to select everything except one condition, it is better to use the opposite. We have to remember the formula “!”, this is used in logical values to get the opposite:

```

Subset <- Data |> filter(!Condition %in% "SEB")
Subset |> pull(Condition) |> unique()

```

[1] “Ctrl” “PPD”

we can use “!” also with the select() function:

```

Subset <- Data |> select(!timepoint)
Subset[1:3,]

```

bid	Condition	Date	infant_sex	pctype	root	singletsFSC	singletsSSC	singletsSSCB
1	INF0052 Ctrl	2025-07-26	Male	HEU-hi	2098368	1894070	1666179	1537396
0.8820349	0.8627649	2	INF0100 Ctrl	2025-07-26	Male	HEU-lo	2020184	1791890
1579098	0.9106762	0.9052256	0.8602660	3	INF0100 Ctrl	2025-07-26	Male	HEU-lo
1033320	875465	845446	0.9705765	0.9845400	0.9578793	lymphocytes live Dump+ Dump- Tcells		
Vd2+ Vd2- Va7.2+ Va7.2- CD4+ CD4- CD8+ 1	0.6420138	0.9020581	0.21090996	0.6911482				
0.2804264	0.008120361	0.9918796	0.01448070	0.9773989	0.6341164	0.3432825	0.2734826	2
0.2145848	0.8908981	0.06252775	0.8283703	0.6748298	0.007265620	0.9927344	0.01577499	
0.9769594	0.6119112	0.3650482	0.3357696	3	0.7403110	0.8757665	0.20023803	0.6755285
0.6119129	0.004651313	0.9953487	0.01579402	0.9795547	0.6639621	0.3155925	0.2862104	
CD8- Tcells_count lymphocytes_count Monocytes Debris CD45_count 1	0.06979990	164771						
587573	0.11796509	0.13723513	915203	2	0.02927858	208241	308583	0.09477437
0.13973396								
1438047	3	0.02938209	371723	607477	0.01545999	0.04212072	820570	

We can also use the `mutate()` function if we want to alter the content of a column or add new ones. Let's start by transforming our current `Data` in to a smaller workable `data.frame`, highlighting the functions and pipes we learned about today:

```
TidyData <- Data |> filter(Condition %in% "Ctrl") |> filter(timepoint %in% "0") |> select(bi
TidyData
```

```
specimen Date timepoint condition Tcells_count CD45_count 1 INF0052 2025-07-26 0 Ctrl 164771 915203 2 INF0100 2025-07-26 0 Ctrl 208241 1438047 3 INF0179 2025-07-26 0 Ctrl 291777 940733 4 INF0134 2025-07-29 0 Ctrl 127866 689676 5 INF0148 2025-07-29 0 Ctrl 234335 1013985 6 INF0191 2025-07-29 0 Ctrl 55780 715443 7 INF0124 2025-07-31 0 Ctrl 70297 687720 8 INF0149 2025-07-31 0 Ctrl 107900 857845 9 INF0169 2025-07-31 0 Ctrl 75540 854594 10 INF0019 2025-08-05 0 Ctrl 208055 873622 11 INF0032 2025-08-05 0 Ctrl 361034 753064 12 INF0180 2025-08-05 0 Ctrl 284958 1049663 13 INF0155 2025-08-07 0 Ctrl 281626 1065048 14 INF0158 2025-08-07 0 Ctrl 280913 1249338 15 INF0159 2025-08-07 0 Ctrl 452551 1190219 16 INF0013 2025-08-22 0 Ctrl 182751 836573 17 INF0023 2025-08-22 0 Ctrl 218435 968035 18 INF0030 2025-08-22 0 Ctrl 85521 732321 19 INF0166 2025-08-28 0 Ctrl 225650 739495 20 INF0199 2025-08-28 0 Ctrl 169736 1112176 21 INF0207 2025-08-28 0 Ctrl 39055 905365 22 INF0614 2025-08-30 0 Ctrl 224396 1569007 23 INF0622 2025-08-30 0 Ctrl 161924 939307
```

And now, we can use the `mutate()` function. In this example, we add a new column based on the `Tcells_count` and `CD45_count`:

```
TidyData <- TidyData |> mutate(Tcells_ProportionCD45 = Tcells_count / CD45_count)
TidyData
```

```
specimen Date timepoint condition Tcells_count CD45_count Tcells_ProportionCD45 1 INF0052 2025-07-26 0 Ctrl 164771 915203 0.18003765 2 INF0100 2025-07-26 0 Ctrl 208241 1438047 0.14480820 3 INF0179 2025-07-26 0 Ctrl 291777 940733 0.31015921 4 INF0134 2025-07-29 0 Ctrl 127866 689676 0.18540010 5 INF0148 2025-07-29 0 Ctrl 234335 1013985 0.23110302 6 INF0191 2025-07-29 0 Ctrl 55780 715443 0.07796568 7 INF0124 2025-07-31 0 Ctrl 70297 687720 0.10221747 8 INF0149 2025-07-31 0 Ctrl 107900 857845 0.12578030 9 INF0169 2025-07-31 0 Ctrl 75540 854594 0.08839285 10 INF0019 2025-08-05 0 Ctrl 208055 873622 0.23815220 11 INF0032 2025-08-05 0 Ctrl 361034 753064 0.47942008 12 INF0180 2025-08-05 0 Ctrl 284958 1049663 0.27147570 13 INF0155 2025-08-07 0 Ctrl 281626 1065048 0.26442564 14 INF0158 2025-08-07 0 Ctrl 280913 1249338 0.22484948 15 INF0159 2025-08-07 0 Ctrl 452551 1190219 0.38022498 16 INF0013 2025-08-22 0 Ctrl 182751 836573 0.21845195 17 INF0023 2025-08-22 0 Ctrl 218435 968035 0.22564783 18 INF0030 2025-08-22 0 Ctrl 85521 732321 0.11678076 19 INF0166 2025-08-28 0 Ctrl 225650 739495 0.30514067 20 INF0199 2025-08-28 0 Ctrl 169736 1112176 0.15261613 21 INF0207 2025-08-28 0 Ctrl 39055 905365 0.04313730 22 INF0614 2025-08-30 0 Ctrl 224396 1569007 0.14301785 23 INF0622 2025-08-30 0 Ctrl 161924 939307 0.17238666
```

We can see that we have many significant digits being returned. Let's round this new column to 2 significant digits by applying the `round()` function

```
TidyData <- TidyData |> mutate(TcellsRounded = round(Tcells_ProportionCD45, 2))
TidyData
```

```
specimen Date timepoint condition Tcells_count CD45_count Tcells_ProportionCD45 Tcell-
sRounded 1 INF0052 2025-07-26 0 Ctrl 164771 915203 0.18003765 0.18 2 INF0100 2025-07-26 0
Ctrl 208241 1438047 0.14480820 0.14 3 INF0179 2025-07-26 0 Ctrl 291777 940733 0.31015921
0.31 4 INF0134 2025-07-29 0 Ctrl 127866 689676 0.18540010 0.19 5 INF0148 2025-07-29 0
Ctrl 234335 1013985 0.23110302 0.23 6 INF0191 2025-07-29 0 Ctrl 55780 715443 0.07796568
0.08 7 INF0124 2025-07-31 0 Ctrl 70297 687720 0.10221747 0.10 8 INF0149 2025-07-31 0 Ctrl
107900 857845 0.12578030 0.13 9 INF0169 2025-07-31 0 Ctrl 75540 854594 0.08839285 0.09
10 INF0019 2025-08-05 0 Ctrl 208055 873622 0.23815220 0.24 11 INF0032 2025-08-05 0 Ctrl
361034 753064 0.47942008 0.48 12 INF0180 2025-08-05 0 Ctrl 284958 1049663 0.27147570 0.27
13 INF0155 2025-08-07 0 Ctrl 281626 1065048 0.26442564 0.26 14 INF0158 2025-08-07 0 Ctrl
280913 1249338 0.22484948 0.22 15 INF0159 2025-08-07 0 Ctrl 452551 1190219 0.38022498
0.38 16 INF0013 2025-08-22 0 Ctrl 182751 836573 0.21845195 0.22 17 INF0023 2025-08-22 0
Ctrl 218435 968035 0.22564783 0.23 18 INF0030 2025-08-22 0 Ctrl 85521 732321 0.11678076
0.12 19 INF0166 2025-08-28 0 Ctrl 225650 739495 0.30514067 0.31 20 INF0199 2025-08-28 0
Ctrl 169736 1112176 0.15261613 0.15 21 INF0207 2025-08-28 0 Ctrl 39055 905365 0.04313730
0.04 22 INF0614 2025-08-30 0 Ctrl 224396 1569007 0.14301785 0.14 23 INF0622 2025-08-30 0
Ctrl 161924 939307 0.17238666 0.17
```

Let's rearrange the rows so that they are descending based on the Tcell proportion. We can use this by using the `desc()` and `arrange()` functions from `dplyr`:

```
TidyData <- TidyData |> arrange(desc(TcellsRounded))
TidyData
```

```
specimen Date timepoint condition Tcells_count CD45_count Tcells_ProportionCD45 Tcell-
sRounded 1 INF0032 2025-08-05 0 Ctrl 361034 753064 0.47942008 0.48 2 INF0159 2025-08-07
0 Ctrl 452551 1190219 0.38022498 0.38 3 INF0179 2025-07-26 0 Ctrl 291777 940733 0.31015921
0.31 4 INF0166 2025-08-28 0 Ctrl 225650 739495 0.30514067 0.31 5 INF0180 2025-08-05 0 Ctrl
284958 1049663 0.27147570 0.27 6 INF0155 2025-08-07 0 Ctrl 281626 1065048 0.26442564 0.26
7 INF0019 2025-08-05 0 Ctrl 208055 873622 0.23815220 0.24 8 INF0148 2025-07-29 0 Ctrl
234335 1013985 0.23110302 0.23 9 INF0023 2025-08-22 0 Ctrl 218435 968035 0.22564783 0.23
10 INF0158 2025-08-07 0 Ctrl 280913 1249338 0.22484948 0.22 11 INF0013 2025-08-22 0 Ctrl
182751 836573 0.21845195 0.22 12 INF0134 2025-07-29 0 Ctrl 127866 689676 0.18540010 0.19
13 INF0052 2025-07-26 0 Ctrl 164771 915203 0.18003765 0.18 14 INF0622 2025-08-30 0 Ctrl
161924 939307 0.17238666 0.17 15 INF0199 2025-08-28 0 Ctrl 169736 1112176 0.15261613 0.15
16 INF0100 2025-07-26 0 Ctrl 208241 1438047 0.14480820 0.14 17 INF0614 2025-08-30 0 Ctrl
```

```
224396 1569007 0.14301785 0.14 18 INF0149 2025-07-31 0 Ctrl 107900 857845 0.12578030 0.13
19 INF0030 2025-08-22 0 Ctrl 85521 732321 0.11678076 0.12 20 INF0124 2025-07-31 0 Ctrl
70297 687720 0.10221747 0.10 21 INF0169 2025-07-31 0 Ctrl 75540 854594 0.08839285 0.09 22
INF0191 2025-07-29 0 Ctrl 55780 715443 0.07796568 0.08 23 INF0207 2025-08-28 0 Ctrl 39055
905365 0.04313730 0.04
```

And let's go ahead and `filter()` and identify the specimens that had more than 30% T cells as part of the overall CD45 gate (context, these samples were Cord Blood):

```
TidyData |> filter(TcellsRounded > 0.3)
```

```
specimen Date timepoint condition Tcells_count CD45_count Tcells_ProportionCD45 Tcell-
sRounded 1 INF0032 2025-08-05 0 Ctrl 361034 753064 0.4794201 0.48 2 INF0159 2025-08-07
0 Ctrl 452551 1190219 0.3802250 0.38 3 INF0179 2025-07-26 0 Ctrl 291777 940733 0.3101592
0.31 4 INF0166 2025-08-28 0 Ctrl 225650 739495 0.3051407 0.31
```

If we want just the ID (specimen) from this last list of items, we can use the `pull()` function:

```
TidyData |> filter(TcellsRounded > 0.3) |> pull(specimen)
```

```
[1] "INF0032" "INF0159" "INF0179" "INF0166"
```

And finally, since I may want to send the data to a supervisor, let's go ahead and export this "tidyed" version of our `data.frame` out to its own `.csv` file. Working within our project folder, this would look like this:

```
NewName <- paste0("MyNewDataset", ".csv")
StorageLocation <- file.path("C:/Users/Macarena/CytometryInR/course/week04/04_IntroToTidyverse/
StorageLocation")
```

```
[1] "C:/Users/Macarena/CytometryInR/course/week04/04_IntroToTidyverse/data/MyNew-
Dataset.csv"
```

```
write.csv(TidyData, StorageLocation, row.names=FALSE)
```